Chapter 20

Measurement Methods, Systems, Capabilities, and Data

MEASUREMENT METHODS

Measurements are made by using one or more well-defined measurement methods to obtain quantifiable information about an object or phenomena. These measurement methods include certain fundamental characteristics that allow them to be used to categorize different types of measurements. Usually, a particular situation or application will make obvious the appropriate measurement method(s) needed to achieve the desired measurement results. Knowledge and experience tell us which specific measurement method will typically yield the best results for a particular situation or application. Factors such as phenomena stability, resolution requirements, environmental influences, timing restraints, and so on, must be considered to determine the optimum measurement method(s). Understanding the mechanics and theory behind measurement methods is helpful not only for determining the best method for a particular situation or application, but also for understanding its limitations and the measurement results it produces.

Measurement methods employ various metrology-related terms that are useful when interpreting the method. Table 20.1 lists definitions from the *VIM*, including their respective *VIM* reference number, for some of these frequently used terms:

Table 20.1 Frequently used VIM terms.[1]

Metrology term	VIM definition
Kind of quantity (1.2)	Aspect common to mutually comparable quantities
Measurement (2.1)	Process of experimentally obtaining one or more *quantity values* that can reasonably be attributed to a *quantity*
Measurand (2.3)	*Quantity* intended to be measured
Method of measurement (2.5)	Generic description of a logical organization of operations used in a measurement
Quantity (1.1)	Property of a phenomenon, body, or substance, where the property has a magnitude that can be expressed as a number and reference
Quantity value (1.19)	Number and reference expressing magnitude of a *quantity*
Reference quantity value (5.18)	*Quantity value* used as a basis for comparison with values of *quantities* of the same *kind*

Table 20.3 *Continued*

Measurement method	Examples
Reciprocity	Determining the sensitivity of a microphone via the response of another microphone
Substitution	Measuring weight using a single-pan scale
Transfer	Determining VAC using an AC/DC transfer device, determining timing deviations via a portable synchronized quartz clock

Measurement methods are the schemes by which measurement data are obtained. The type of measurement method that is appropriate for a particular situation should be thought out and determined before engaging in a measurement. Selection of an inappropriate measurement method will result in wasted time and other resources and produce undesirable and/or unreliable measurement data. Looking before leaping is sound advice when it comes to selecting a measurement method.

MEASUREMENT SYSTEMS

Measurement systems are how measurement data are obtained. A measurement system is an ensemble comprising various elements such as measurement personnel, calibration standards, measurement devices, measurement fixtures, measurement environment, measurement methodology, and so on. These systems are used to obtain quantifiable, attributable data related to an object or phenomenon. You can visualize a measurement system as a process of interactive, interrelated activities by which various objects or phenomena are related to measurement data.

Measurement systems are created and used based on needs for specific measurement data. The makeup of a measurement system is determined by an application or particular situation. The adequacy of a measurement system depends on the accuracy and reliability requirements of the measurement data. Less stringent requirements demand less of a measurement system in terms of sophistication, variability, repeatability, and so on. How the measurement data will be used will drive the selection, composition, and sophistication of a measurement system in order to meet measurement objectives.

Measurement systems are designed to produce measurement data that are assumed to be faithful and representative of the measurand(s) they are intended to measure. Table 20.4 lists *VIM* definitions, including their respective *VIM* reference numbers, that are applicable to this discussion.

For a measurement system to be properly constructed, a comprehensive understanding of applicable measurement application(s) is required. This understanding will direct the development and definition of the measurement system. Also, a thorough understanding of measurement data requirements will provide guidance as to the confidence level and reliability traits required of the measurement system. NASA Reference Publication 1342, *Metrology—Calibration*

interval (see Chapter 10) or in the estimation that measurements may be expected to lie within a probable range (see Chapter 29).

Calibration considerations regarding measurement systems are of particular interest to metrologists. NASA Reference Publication 1342 notes that "[m]easurement processes are accompanied by errors and uncertainties that cannot be eliminated. However, they can be quantified and limited or controlled to 'acceptable' levels. Calibration is done for this purpose."[6]

Fluke's *Calibration: Philosophy in Practice* notes that calibration is "[a] set of operations, performed in accordance with a definite, documented procedure, that compares the measurements performed by an instrument to those made by a more accurate instrument or standard, for the purpose of detecting and reporting, or eliminating by adjustment, errors in the instrument tested."[7] Calibration can thus be considered as defining attributes about a measurement system in terms of how well they can correlate an unknown (object or phenomenon being measured) to a known (calibration standard). Calibration relates measurement systems to performance indices, thereby providing the means to estimate expected performance.

Measurement systems, like manufacturing processes, transform inputs to desired outputs and are composed of various interactive, interdependent elements that determine the quality and makeup of these outputs. Careful consideration must be given to the development or selection of a measurement system in order for it to correspond with measurement application requirements, thus ensuring the validity and usability of derived measurement data. Measurements are metrological tools used in performing measurement tasks. As the age-old adage says, "The right tool for the right job" cannot be overemphasized.

MEASUREMENT CAPABILITIES

Measurement systems embody a variety of measurement capabilities inherent in their design for their intended purpose(s). *Measurement capabilities* are attributes of a measurement system that determine the extent to which measurements may be made within some qualifying restraints such as measurement range, ambient conditions, required input amplitude, and so on. Measurement system capabilities should be congruent with the requirements of the measurement application they are intended for. Whether a measurement system has the required capabilities to meet a measurement application is not always readily apparent and often must be established through user-assessment activities. It is the responsibility of the measurement system user (or their management) to ascertain whether a measurement system is capable of meeting the requirements of a particular measurement application. Intentional/unintentional use of a measurement system in terms of operation beyond its established capabilities normally results in measurement data with unknown uncertainties at best, or completely erroneous measurement data in worst-case scenarios.

The adequacy of a measurement system to fulfill the requirements of a measurement application is addressed within ISO 9001:2015, Clause 7.1.5, "Monitoring and measuring resources," sub-clause 7.1.5.1, General: "The organization shall determine and provide the resources needed to ensure valid and reliable results when monitoring and measuring is used to verify the conformity

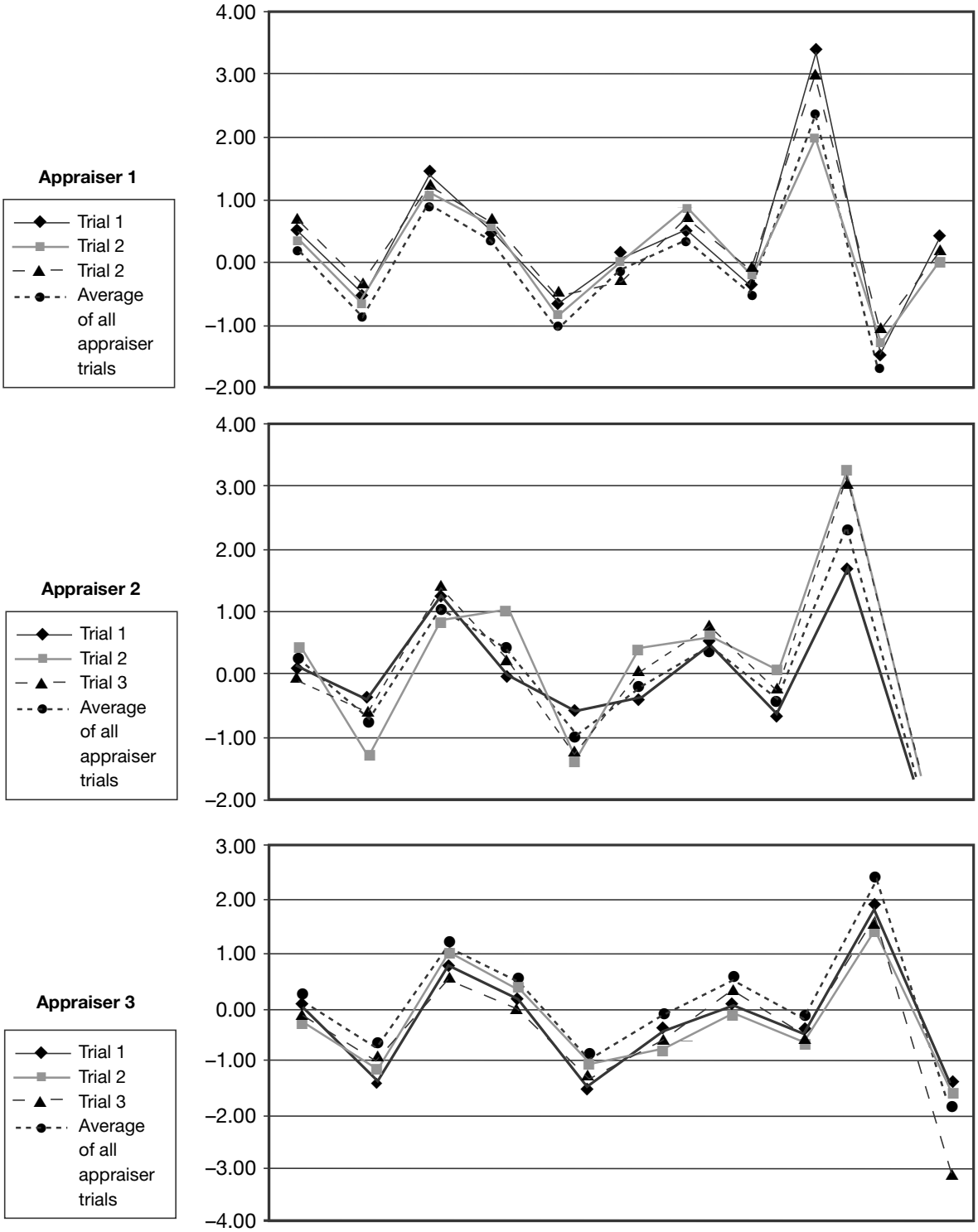


Figure 20.4 Example of appraiser variations.

MEASUREMENT DATA

A measurement system's principal purpose is to generate measurement data. Measurement data come in many varieties, such as alphanumeric characters, plots and graphs, increasing or decreasing audiovisual displays, limit indicators, and so on. Measurement data type should be compatible with the requirements of the intended measurement application. Measurement data are often the only basis

- Being interpreted outside of their intended context
- Disclosing capabilities or limitations
- Giving insight into programs or products in development

Measurement data confidentiality should be explicitly addressed, and proper safeguards incorporated to prevent unauthorized disclosure.

Measurement data considerations can, if not satisfactorily addressed, make good data unusable for their intended purpose(s). Without a satisfactory understanding of measurement systems output data in relation to the requirements of a measurement application, said data consideration can result in time and effort being wasted and bad decisions being made. Measurement practitioners and those interpreting measurement data would do well to consider how measurement data are to be used before selecting a measurement system for a measurement application to avoid many of the pitfalls.

CALIBRATION METHODS AND TECHNIQUES

IM&TE requires the use of various methods and techniques for determining whether a unit is operating to its published specifications. These methods and techniques are used to establish a relationship between an applied signal and the corresponding IM&TE measurement display, cancel out residuals that can offset the measurand of interest, provide sufficient sensitivity to determine small differences between a unit and a calibration standard, and so on. Often, a calibration method/technique is recommended by the original equipment manufacturer (OEM) as a necessary step to be performed prior to using a unit.

The selection of a particular calibration method/technique is dictated by various factors. These factors include the measurand of interest, the inherent functionality and/or limitations of the IM&TE, the measurement scenario and its associated environment, the operator knowledge and skill, OEM recommendations, and so on. Use of an inappropriate calibration method/technique can often mask an IM&TE measurement response, rendering it inaccurate. Not performing an OEM-recommended calibration method/technique can also degrade a unit's performance. It is essential that calibration practitioners be aware of the calibration methods/techniques appropriate for a particular application in order to ensure that their evaluations are based on reliable measurement data and to avoid misadjusting a unit as a result of misleading measurement data.

Calibration methods and techniques can be adapted to a wide variety of IM&TE types. Some of the most common calibration methods and techniques include:

- **Linearization.** This is a method by which IM&TE is corrected for a linear response such that a step change in an applied signal will result in a corresponding step change in the IM&TE indication. This method may also be used to correct a nonlinear output using a linear-responding measuring device. Linearization is typically used to correct nonlinear measurement sensors such as those used to make high-temperature measurements.